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Rational Design of a $LiNbO_3$-like Nonlinear Optical Crystal, Li_2ZrTeO_6, with High Laser-Damage Threshold and Wide Mid-IR Transparency Window

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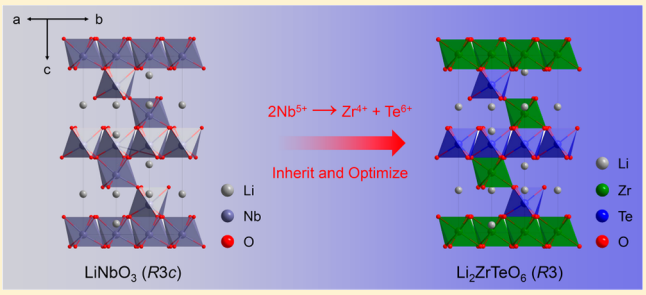
Supporting Information

ABSTRACT: With existing and emerging technologies urgently demanding the expansion of the laser wavelengths, high-performance nonlinear optical (NLO) crystals are becoming indispensable. Here, a prospective NLO crystal, Li_2ZrTeO_6, is rationally designed by the element substitution of Nb for Zr and Te from $LiNbO_3$, which has been recognized as one of the most commercial NLO crystals. Li_2ZrTeO_6 with *R*3 symmetry inherits the structural merits of $LiNbO_3$ (space group *R*3*c*) and thus meets the requirements for NLO applications, including noncentrosymmetric crystal structure, moderate birefringence, and phase-matchability. Moreover, it can be exploited to achieve more outstanding optical damage resistant behavior (>1.3 GW cm^{-2}), exceeding 22 times that of $LiNbO_3$, which is more suitable for high-energy laser applications. Notably, this compound displays the widest IR absorption edge (7.4 μm) among all of the noncentrosymmetric tellurates reported so far. These excellent attributes suggest that Li_2ZrTeO_6 is a promising candidate for providing high NLO performance. The substitution of Nb for Zr and Te from $LiNbO_3$ demonstrates a viable strategy toward the rational design of NLO crystals with anticipated properties.



INTRODUCTION

Nonlinear optical (NLO) crystals, which are capable of generating coherent light through second-harmonic generation (SHG) methods, have garnered plenty of attention in laser science and technology.[1−3] Many efforts have been made over the past decades to elucidate the relationship of the compositions, structures, and NLO characteristics of crystalline compounds; unfortunately, the efficient fabrication of novel NLO crystals with outstanding overall properties, including relatively large SHG response, moderate birefringence, phase-matching capability, high laser-damage threshold, wide transparency window at wavelengths of interest, good thermal stability, and facile growth of large crystals, still remains challenging.[4]

In order to design and construct novel NLO crystals, the conventional approach is suggested to introduce NLO-active structural chromophores, such as second-order Jahn−Teller (SOJT) active cations (octahedrally coordinated d^0 transition-metal cations (Zr^{4+}, Ta^{5+}, Mo^{6+}, etc.) or stereochemically active lone-pair cations (Sn^{2+}, Bi^{3+}, Te^{4+}, etc.)),[5−9] coplanar π-conjugated units ($[BO_3]^{3-}$, $[B_3O_6]^{3-}$, $[B_3O_7]^{5-}$, etc.),[10−17] and polar chalcogenide groups ($[TeS_3]^{2-}$, $[AsS_3]^{3-}$, $[SbS_3]^{3-}$, etc.).[18−22] Among the known materials, $LiNbO_3$, containing SOJT-active Nb^{5+} cations, has been considered as one of the most commercial crystals for applications in nonlinear optics owing to the extraordinary physical properties, such as large NLO coefficients, moderate birefringence, phase-matchability, and high crystal quality.[23] In spite of the distinguished characteristics, the relative low damage threshold of $LiNbO_3$ against photorefractive effects has restricted practical high-power conditions with large device aperture.[24] Moreover, single crystals of $LiNbO_3$ cannot fabricate the NLO devices operating at wavelengths beyond 5 μm, due to the limited optical transparency window (0.35−5 μm).[25] Taking into account the multifunctional effect, Zr^{4+} cations with d^0 electronic configuration not only offer structural distortions[7] but also promote optical damage resistant behavior.[26] In addition, Te^{6+} cations as heavy element ions with d^{10} electrons can construct distorted TeO_6 polyhedra with polar displacement and extend the accessible wavelength range substantially.[27−30]

Guided by these ideas, we focused on a $LiNbO_3$-like NLO crystal by substituting Nb^{5+} cations of $LiNbO_3$ for Zr^{4+} and Te^{6+} cations to maintain the favorable structure of $LiNbO_3$ in addition to large laser-damage threshold and wide transparency

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window, namely, Li_2ZrTeO_6. In 1988, relying on X-ray powder diffraction intensities, it was Choisnet and co-workers who originally worked out the trigonal structure of Li_2ZrTeO_6.[31] Thus far, research has focused exclusively on the polycrystalline forms of Li_2ZrTeO_6.[31−33] Therefore, to obtain the intrinsic properties in the absence of grain boundary effects and to further gain insights into the relationship between the structural parameters and the functional physical characteristics, single crystals of Li_2ZrTeO_6 with high crystalline quality are extremely desired for the acquisition of reliable results.

Herein, we present the synthesis, crystal growth, Laue back-reflection patterns, optical property characterization, and structure−property relationship of a new $LiNbO_3$-like NLO crystal, Li_2ZrTeO_6. The experimental results and theoretical calculations suggest that Li_2ZrTeO_6 is an adequate candidate for mid-IR NLO applications.

■ EXPERIMENTAL SECTION

Synthesis. As raw materials, Li_2CO_3 (99.99%, Alfa-Aesar), ZrO_2 (99.99%, Alfa-Aesar), and TeO_2 (99.99%, Alfa-Aesar) were all used as received. Polycrystalline samples of Li_2ZrTeO_6 were synthesized through the conventional solid-state reaction. Stoichiometric amounts of Li_2CO_3, ZrO_2, and TeO_2 were adequately ground in an agate mortar and then pressed into a pellet. The pellet was transferred into a platinum crucible and then sintered in a programmable muffle furnace to 600 °C gradually and maintained for 80 h with several intermittent regrindings.

Powder X-ray Diffraction (PXRD). The PXRD pattern was recorded employing a Bruker-AXS D8 Advance X-ray diffractometer set for Cu Kα radiation, in the angular range (2θ) from 10° to 90°, with a scanning step time of 0.4 s and a scanning step size of 0.02° at room temperature.

Thermal Stability. Thermogravimetric analysis (TGA) and differential scanning calorimetry (DSC) of ground Li_2ZrTeO_6 polycrystals were simultaneously performed up to 1200 °C at a constant heating rate of 10 °C min^{-1} under a N_2 atmosphere using a TGA/DSC1/1600HT apparatus (Mettler Toledo Instruments).

Single-Crystal Growth. Small crystals of Li_2ZrTeO_6 were obtained through the spontaneous crystallization method with an excess amount of Li_2CO_3 and TeO_2 serving as flux. A reaction mixture of Li_2CO_3, ZrO_2, and TeO_2 at the molar ratio of 2:1:5 was adequately ground and subsequently placed in a platinum crucible. The crucible containing the mixture was gradually heated to 760 °C and held for 2 days to ensure a completely homogeneous solution at the very center of a programmable-temperature electric furnace. A platinum wire was slowly dipped into the solution to serve as a nucleation center, and then the temperature of the solution was decreased to 692 °C at a rate of 2 °C h^{-1}. Some colorless and transparent small crystals were obtained on the platinum wire by spontaneous nucleation.

Single crystals of Li_2ZrTeO_6 were obtained through the top seeded solution growth (TSSG) technique. The saturation temperature of the solution was determined accurately by observing the dissolution or growth of a seed crystal dipped into the solution. A high-quality Li_2ZrTeO_6 seed was introduced slowly into the solution at a temperature 4 °C above the saturation temperature to dissolve the surface defects of the seed. Subsequently, a slow cooling rate of about 0.09 °C h^{-1} was applied to begin the growth process until the centimeter-sized crystal was obtained. When the process was completed, the crystal was lifted out of the surface of the high-temperature solution and slowly cooled to ambient temperature within 3 days.

Single-Crystal X-ray Diffraction. A transparent block-shaped single crystal of Li_2ZrTeO_6 with dimensions of 0.116 × 0.133 × 0.109 mm^3 was selected and affixed on the tip of a thin glass fiber for the single-crystal structure determination. The diffraction data collection was carried out on a Bruker AXS SMART diffractometer with monochromatic Mo Kα radiation (λ = 0.71073 Å) at room temperature.[34] The data integration and unit cell refinement were determined by the Siemens SAINT program of the standard APEX II software,[34] and absorption corrections were acquired according to a multiscan-type model.[34] The initial model of the structure was obtained by the direct method. Subsequent difference Fourier maps and full-matrix least-squares refinement fitting on F_0^2 were performed using the SHELXTL crystallographic software package.[35] The details of the crystal structure (CCDC 1861704) can be obtained from the Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.

Laue Back-Reflection Measurement. In order to assess the crystalline perfection of the harvested Li_2ZrTeO_6 single crystal, the Laue back-reflection measurement was carried out on a real-time back-reflection Laue camera system (Multiwire MWL 120 with Northstar software). A (012) oriented slice of Li_2ZrTeO_6, mechanically polished on both sides, whose dimensions are 4 × 4 × 1 mm^3, was employed for the Laue diffraction measurement.

UV−Visible Diffuse Reflectance Spectra. UV−visible diffuse reflectance data for ground Li_2ZrTeO_6 polycrystals were obtained over the wavelength range extended from 200 to 900 nm with a Shimadzu UV 2550 spectrophotometer using an integrating sphere, while $BaSO_4$ (a powder sample) was employed as a reference material for baseline correction. On the basis of the Kubelka−Munk equation,[36] reflectance spectra were transformed to relative absorbance.

Laser-Damage Threshold Measurement. A well-polished crystal plate of Li_2ZrTeO_6 with a size of 6 × 6 × 2 mm^3 was employed for the determination of laser-damage threshold. The measurement was performed on a diode-pumped Nd:YAG nanosecond laser (Minilite II, Continuum) at a laser wavelength of 1064 nm with a pulse width of 10 ns. The pulse energy was maintained to around 35 mJ, and the pulse repetition rate was 10 Hz. A convex lens was set in front of the sample to focus the laser beam. A $LiNbO_3$ crystal wafer was used as a benchmark in the entire procedure.

Transmission Spectroscopy. The optical transmission spectra of a Li_2ZrTeO_6 single crystal were collected on a Hitachi U-3500 UV−vis−IR spectrometer in the 200−2000 nm region and a Nicolet NEXUS 670 FTIR spectrometer in the 2000−8000 nm region, respectively. One square piece of well-polished crystal wafer with a size of 4 × 4 × 1 mm^3 was used to perform the measurements.

Second-Harmonic Generation. The SHG response was measured at room temperature according to the method proposed by Kurtz and Perry with a diode-pumped Nd:YAG laser operating at an optical wavelength of 1064 nm.[37] The powder SHG efficiency has been indicated to be closely related to the particle size;[37] thus the ground powders of the Li_2ZrTeO_6 crystal were graded into the following several different particle size ranges: <50, 50−75, 75−107, 107−150, 150−250, 250−300, and 300−375 μm. The microcrystalline KH_2PO_4 (KDP) as a reference was graded into the corresponding size ranges to make pertinent comparisons.

Computational Details. First-principles calculations were carried out utilizing the plane-wave pseudopotential method implemented in the CASTEP code through density functional theory (DFT).[38,39] Calculations on the structure of Li_2ZrTeO_6 used the experimental cell parameters and atomic positions. The exchange−correlation potential was described by adopting the generalized gradient approximation (GGA) form in the scheme of the Perdew−Burke−Eruzerhof for solids (PBEsol) functional.[40] The interactions between the ionic cores and the valence electrons for all constituent elements were simulated by the norm-conserving pseudopotential.[41] Li $2s^1$, Zr $4s^2$ $4p^6$ $4d^2$ $5s^2$, Te $5s^2$ $5p^4$, and O $2s^2$ $2p^4$ were treated as valence electrons. The kinetic energy cutoff for the plane-wave basis was set to be 750 eV, and the numerical integration of the Brillouin zone was carried out utilizing a 6 × 3 × 4 Monkhorst−Pack k-point grid. During the optical properties calculations, more than 100 empty bands were used to guarantee high accuracy. The optical refractive indices n and birefringence Δn can be determined based on the calculated electronic transition between the unoccupied and occupied states caused by the interactions between the photons.[42] The default values of the CASTEP code were kept, along with the other parameters used in the calculations and convergent criteria.

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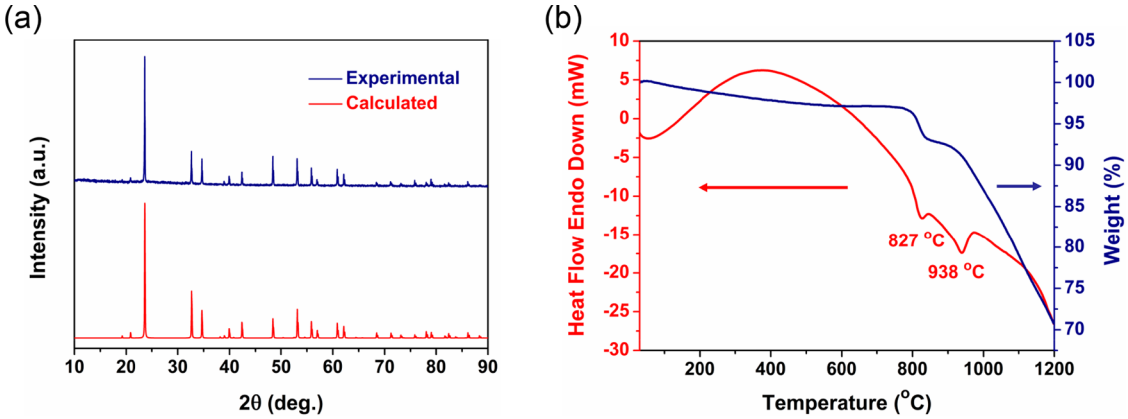

Figure 1. (a) Experimental and calculated PXRD patterns for polycrystalline Li_2ZrTeO_6 (phase-pure). (b) TGA and DSC curves for polycrystalline Li_2ZrTeO_6.

RESULTS AND DISCUSSION

Synthesis and Thermal Behavior. Polycrystalline samples of Li_2ZrTeO_6 were prepared from the stoichiometric mixture of Li_2CO_3, ZrO_2, and TeO_2 through the conventional solid-state reaction. The PXRD pattern on the white polycrystalline samples (see Figure 1a) showed that the prepared material was phase-pure and consistent with the theoretical pattern from the single-crystal structural data. The thermal stability of Li_2ZrTeO_6 was evaluated by the TGA and DSC measurements. As shown in Figure 1b, there are two drastic endothermic peaks around 827 and 938 °C in the DSC curve; meanwhile mass losses are observed in the TGA curve. The PXRD detection of the solid residue after heating at 850 °C for 30 min revealed that Li_2ZrTeO_6 decomposed to Li_4ZrO_4. This evidence indicates that Li_2ZrTeO_6 is an incongruent melting compound. Accordingly, a high-temperature flux method has to be adopted to obtain large-size Li_2ZrTeO_6 crystals below the decomposition temperature.

Crystal Growth. Due to the incongruent melting nature of Li_2ZrTeO_6, it must be grown below the decomposition temperature by the flux method. The selection of a suitable flux is vital for the obtainment of high-quality crystals. Because of the conveniently low melting point and the superior ability to dissolve oxides, TeO_2 as a flux has been widely utilized to grow single crystals of a large variety of oxides containing tellurium element.[43−47] In addition, a special effect of Li^+ cations substantially enlarges the dimensions of grown crystals.[48] Therefore, we have sought to combine TeO_2 and Li_2CO_3 to serve as flux. As shown in Figure S1 of the Supporting Information, a centimeter-sized Li_2ZrTeO_6 single crystal with dimensions of 16 × 15 × 12 mm^3 was successfully harvested through the TSSG technique in 10 days. The environmentally friendly habits and the simplicity of the crystal growth (neither toxic ingredients introduced nor rigorous atmosphere needed) are particularly advantageous to the practical NLO applications of the Li_2ZrTeO_6 crystal.

Crystal Structure of Li_2ZrTeO_6. Li_2ZrTeO_6 exhibits a structure that closely resembles that of the polar oxide $LiNbO_3$.[49,50] The crystal structure of Li_2ZrTeO_6 belongs to a trigonal crystal system with a polar space group of *R*3 (No. 146). The refined crystallographic data and details of the experimental conditions for Li_2ZrTeO_6 are given in Table 1. Figure 2 shows the crystal structures of $LiNbO_3$ and Li_2ZrTeO_6 from the perspective of a polyhedral geometry. In Li_2ZrTeO_6 the Zr and Te atoms are located at the octahedral sites, which are occupied by Nb atoms in $LiNbO_3$. The cation ordering sequence is Zr−Li−Te−Li−Zr... along the *c*-axis (Figure 2). As expected for the $LiNbO_3$-like structure, the octahedra share corners to construct the three-dimensional framework holding the structure firmly together. As shown in Figure 3, the Zr^{4+} cations are located in a distorted octahedral environment with the Zr−O distances ranging from 2.070(8) to 2.110(7) Å, which originate from the SOJT effect.[7] Meanwhile, the Te^{6+} cations with d^{10} electronic configuration with polar displacement are also bonded to six O atoms in distorted TeO_6 octahedra with Te−O bond lengths ranging from 1.909(7) to 1.916(8) Å. The detailed crystallographic data of Li_2ZrTeO_6 are given in Tables S1−S3 of the Supporting Information. The intraoctahedral distortion parameters can be determined according to the following expression: $\Delta = 1/6 \sum_i[(d_i - d_{av})/d_{av}]^2$, in which d_i and d_{av} represent the individual bond length and the average bond length, respectively.[51] The Δ for ZrO_6 octahedra in Li_2ZrTeO_6 is 9.11 × 10^{-5}, which is about 30 times larger than that of TeO_6 octahedra (Δ = 3.44 × 10^{-6}). Obviously, ZrO_6

Table 1. Crystal Data and Structure Refinement for Li_2ZrTeO_6

empirical formula	Li_2ZrTeO_6
fw	328.70
temperature (K)	296(2)
wavelength (Å)	0.71073
cryst syst	trigonal
space group	*R*3 (No. 146)
a (Å)	5.175(2)
c (Å)	13.857(6)
V (Å^3)	321.4(3)
Z	3
ρ_{calcd} (g cm^{-3})	5.095
μ (mm^{-1})	9.176
F(000)	438
cryst size (mm^3)	0.116 × 0.133 × 0.109
data/restraints/params	320/1/25
R(int)	0.0209
GOF (F^2)	1.091
final *R* indices [$I > 2\sigma(I)$][a]	R_1 = 0.0140, wR_2 = 0.0357
R indices (all data)	R_1 = 0.0140, wR_2 = 0.0357
absolute structure param	0.29(5)

[a] $R_1 = \sum||F_o| - |F_c||/\sum|F_o|$ and $wR_2 = \{\sum[w(|F_o|^2 - |F_c|^2)^2]/\sum[w(|F_o|^4)]\}^{1/2}$.

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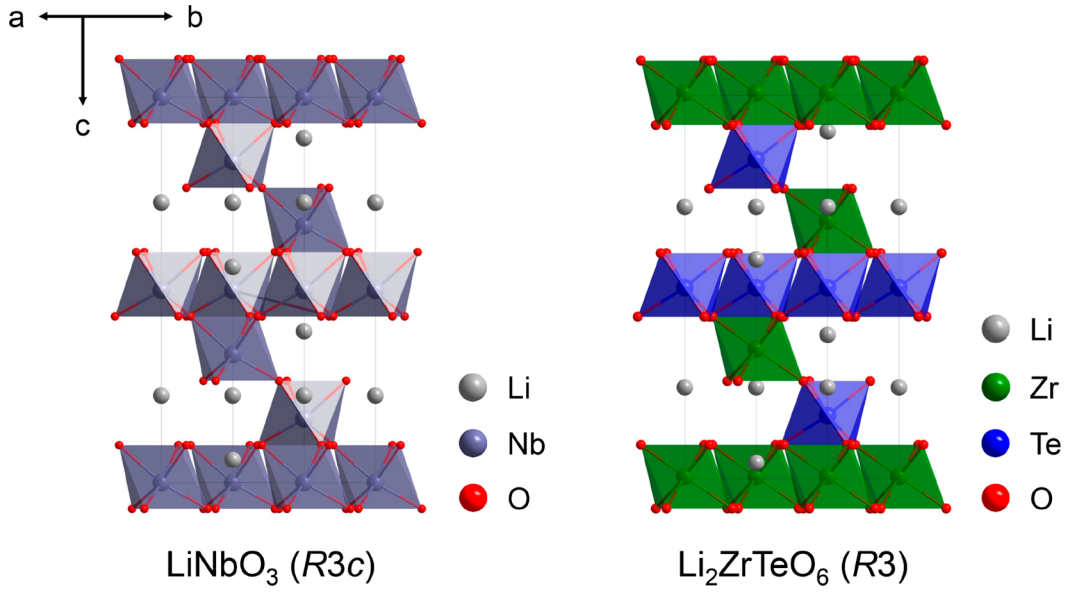

Figure 2. Comparison of crystal structures between $LiNbO_3$ and Li_2ZrTeO_6.

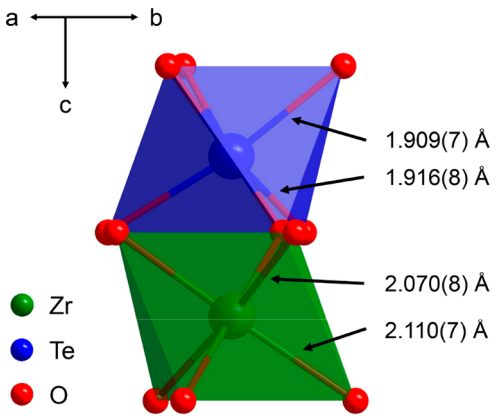

Figure 3. Locally coordinated environment of ZrO_6 and TeO_6 octahedra oriented along the *c*-axis, displaying a corner-sharing octahedral pair.

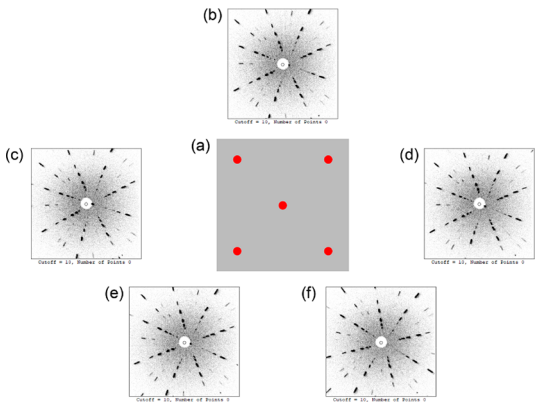

Figure 4. (a) Schematic positions (red dots) of the Laue back-reflection measurement on the crystal wafer. (b−f) From left to right and up to down, characteristic Laue back-reflection patterns at different positions with the X-ray beam hitting the crystal wafer.

octahedra exhibit more considerable distortion as a result of off-centering of the Zr^{4+} cations, unlike relatively regular TeO_6 octahedra. The crystal structure obtained is consistent with the result reported by Choisnet and co-workers.[31]

Laue Back-Reflection Patterns. To facilitate a precise investigation of the crystallinity of an as-grown crystal, the Laue back-reflection measurement is broadly exploited.[52−54] As can be apparently seen from Figure 4, the characteristic Laue back-reflection patterns at different positions of the crystal wafer are uniform, clear, and bright. It demonstrates that the crystalline quality of the Li_2ZrTeO_6 crystal is high enough, which furnishes the basis for performing the measurements and evaluations of intrinsic physical properties.

Optical Properties. Optical damage in NLO crystals can hinder the performance at high laser intensity, as exemplified by the efficiency and durability of the optical devices involving the nonlinear interactions. Thus, high laser-damage threshold is an important factor for NLO crystals. The laser-damage threshold depends on the optical band gap of materials intrinsically.[55] The UV−visible diffuse reflectance spectra of Li_2ZrTeO_6 polycrystals were collected (see Figure 5a), and the absorption data (K/S) were calculated according to the Kubelka−Munk function,[36] $F(R) = K/S = (1 - R)^2/2R$, in which K represents the absorption coefficient, S represents the scattering factor, and R represents the reflectance. As shown in the inset of Figure 5a, the intercept of the linear part of the ascending curve with the energy axis in the K/S versus E plots can be regarded as the optical band gap of Li_2ZrTeO_6. The band gap of Li_2ZrTeO_6 is estimated to be 4.06 eV, which is significantly larger than that of the commercial NLO crystal $LiNbO_3$ (3.11 eV).[56]

The laser-damage threshold of the Li_2ZrTeO_6 crystal was measured at a wavelength of 1064 nm with a pulse width of 10 ns. A bright SHG green light (532 nm) was obviously observed from the crystal, and there was no indication that the green light underwent any degradation after prolonged laser radiation. Li_2ZrTeO_6 has a high laser-damage threshold of >1.3 GW cm^{-2}, exceeding 22 times that of the benchmark $LiNbO_3$ (60 MW cm^{-2}), which is in agreement with the tendency of the band gap. (It is worth mentioning that the threshold of Li_2ZrTeO_6 is constrained to the measurement range of the equipment.) Moreover, as intuitively shown in Figure 5b, Li_2ZrTeO_6 can be comparable to those widely used

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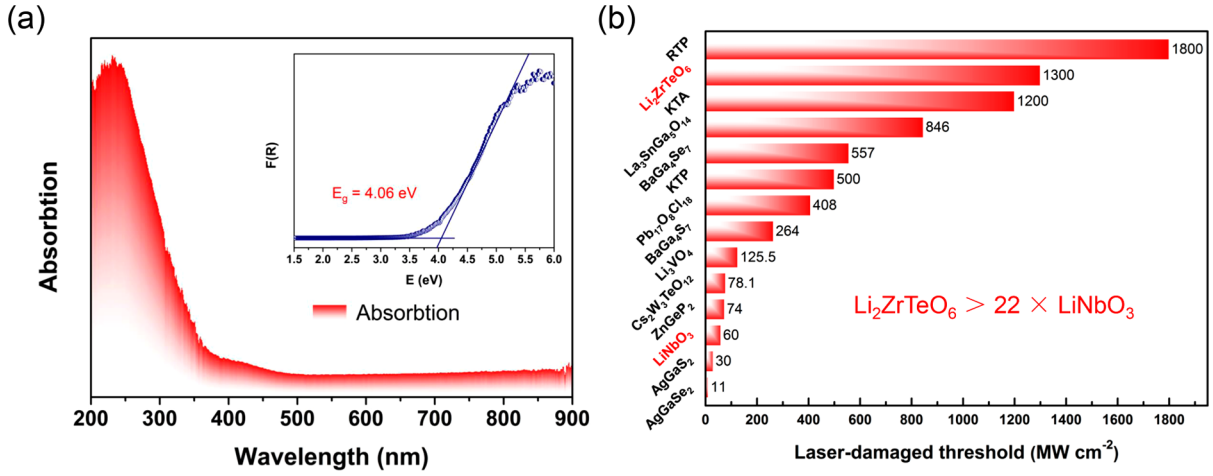

Figure 5. (a) UV−visible diffuse reflectance spectra data for ground powders of the Li_2ZrTeO_6 crystal. The inset shows the relationship between K/S and E (eV). (b) Comparison of laser-damage thresholds among the known mid-IR NLO crystals. (The corresponding measurement details are listed in Table S4.)

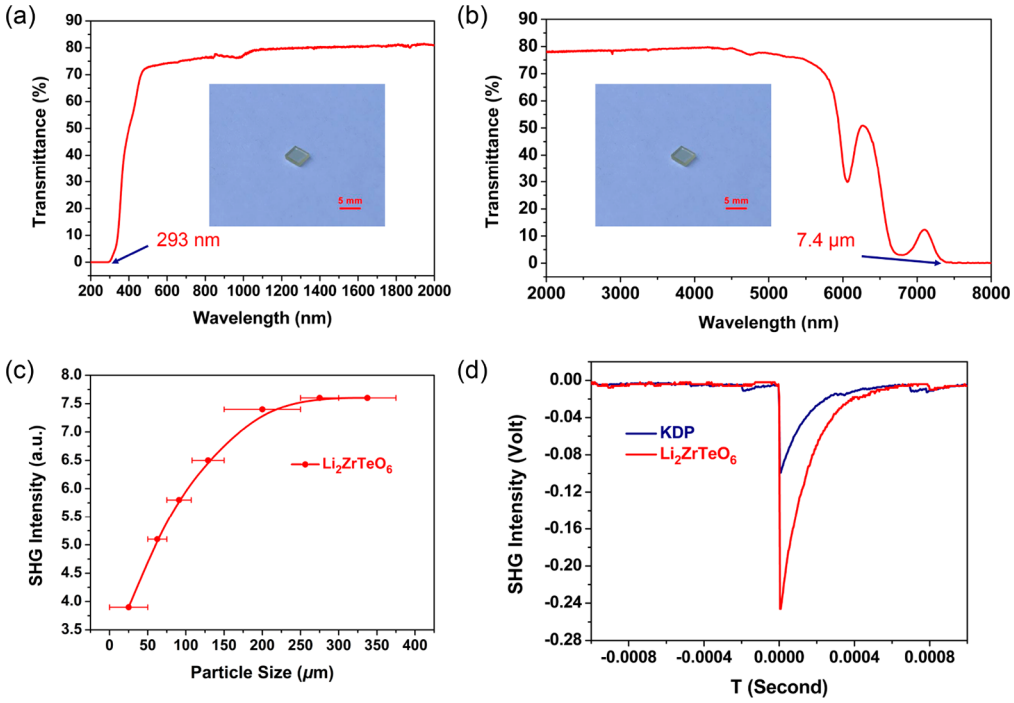

Figure 6. (a) UV−vis and (b) IR transmission spectra of the Li_2ZrTeO_6 single crystal. Inset: Li_2ZrTeO_6 crystal wafer of dimensions 4 × 4 × 1 mm^3. SHG intensity of Li_2ZrTeO_6 with the commercial KDP as a reference: (c) phase-matching curve for Li_2ZrTeO_6 (the red solid curve depicted is to direct the eyes and is not a fit to the data); (d) oscilloscope traces of the SHG signals for Li_2ZrTeO_6 and KDP in the same particle size of 75−107 μm.

mid-IR NLO crystals such as KTP (0.5 GW cm^{-2}),[57] KTA (1.2 GW cm^{-2}),[58] and RTP (1.8 GW cm^{-2}).[59] The results show that Li_2ZrTeO_6 is an excellent candidate suitable for high-power NLO applications.

To obtain an accurate result about the absorption edge, the optical transmission spectra of a Li_2ZrTeO_6 single-crystal sample were recorded in the wavelength range of 200−8000 nm (see Figure 6a,b). The UV absorption edge of the Li_2ZrTeO_6 crystal is located at 293 nm. The Li_2ZrTeO_6 crystal exhibits a high mid-IR transparency up to 6 μm, and the IR absorption edge is extended to 7.4 μm, covering a critical atmospheric transparent window (3−5 μm). To the best of our knowledge, the transparent region of Li_2ZrTeO_6 is the widest among all of the noncentrosymmetric tellurates (see Table 2). These results indicate that Li_2ZrTeO_6 is competent for a large variety of NLO applications in wide wavelength ranges.

Second-Harmonic Generation. Since the compound Li_2ZrTeO_6 crystallizes in the noncentrosymmetric space group $R3$, it is considered to possess an SHG response. The curve of the powder SHG signals detected and recorded is plotted against the particle size in Figure 6c. The SHG intensity of Li_2ZrTeO_6 increases along with the increase of the particle size before it attains the maximum value independent of the particle size, which shows that the compound Li_2ZrTeO_6 is type-I phase-matchable with the incident wavelength at 1064 nm. Moreover, Li_2ZrTeO_6 exhibits an

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Table 2. Comparison of the UV and IR Cutoff Edge among the Noncentrosymmetric Tellurates

compound	crystal system	space group	UV cutoff edge (μm)	IR cutoff edge (μm)
$V_2Te_2O_9$[60]	orthorhombic	*Fdd*2	0.62	6.2
Bi_2TeO_5[61]	orthorhombic	*Abm*2	0.40	7.0
α-$BaMo_2TeO_9$[62]	orthorhombic	$Pca2_1$	0.38	5.5
β-$BaMo_2TeO_9$[63]	monoclinic	$P2_1$	0.4	5.4
$Cs_2W_3TeO_{12}$[30]	hexagonal	$P6_3$	0.41	5.4
$Na_2W_2TeO_9$[64]	monoclinic	*Ia*	0.36	5.8
$CdMoTeO_6$[65]	tetragonal	$P\bar{4}2_1$	0.35	5.4
$Cs_2Mo_3TeO_{12}$[29]	hexagonal	$P6_3$	0.43	5.38
$Na_2Mo_3Te_3O_{16}$[66]	monoclinic	*I*2	0.42	5.4
$MgMoTeO_6$[67]	orthorhombic	$P2_12_12$	0.36	5.2
$MnMoTeO_6$[68]	orthorhombic	$P2_12_12$	0.41	6.25
Zn_2MoTeO_7[69]	monoclinic	$P2_1$	0.3	5.75
Li_2ZrTeO_6 (this work)	trigonal	*R*3	0.29	7.4

adequate SHG efficiency approximately 2.5 times that of the benchmark KDP in the same particle size range (75−107 μm), as shown in Figure 6d.

Structure−Property Relationship. To further gain insights into the microscopic origin of the optical properties of Li_2ZrTeO_6, electronic structure calculations were employed based on DFT.[38,39] As presented in the energy band structure (Figure 7a), Li_2ZrTeO_6 is an indirect band material with an optical band gap of 4.06 eV. (A scissor operation of 0.854 eV has been applied because of the well-known underestimation of the band gap by the GGA method.[70,71]) Meanwhile, Figure 7b shows the detailed bands assigned in accordance with the total and partial densities of states (TDOS and PDOS). Since the optical characteristics of a crystal mostly arise from the electronic transitions close to the Fermi energy,[42] the bottom region of the conduction band (CB) and the upper region of the valence band (VB) are well investigated, from which several features can be deduced: (1) the electron states near the minimum of the CB (4.0 to 5.5 eV) primarily contain Te 5s and O 2p orbitals; (2) the electron states near the maximum of the VB ranging from −4.0 to 0.0 eV are mostly composed of O 2p orbitals, which are hybridized with Zr 4p 4d orbitals. The Te 5p orbitals contribute less to the VB top; (3) the contribution originating from the orbitals of the alkali-metal cations (Li^+) can be neglected in these states. The lowest unoccupied molecular orbital (LUMO) and the highest occupied molecular orbital (HOMO) of Li_2ZrTeO_6 are intuitively visualized in Figure 7c,d. The LUMO in the CB is concentrated on the Te and O atoms, and the HOMO in the VB locates at the O atoms and a little at Zr atoms. According to these analyses, it is ZrO_6 octahedra and TeO_6 octahedra that make a considerable contribution to the optical properties of the Li_2ZrTeO_6 crystal. On the basis of the calculated electronic structure, the dispersion curves of the linear refractive indices of the Li_2ZrTeO_6 crystal are calculated, as showed in Figure 8.

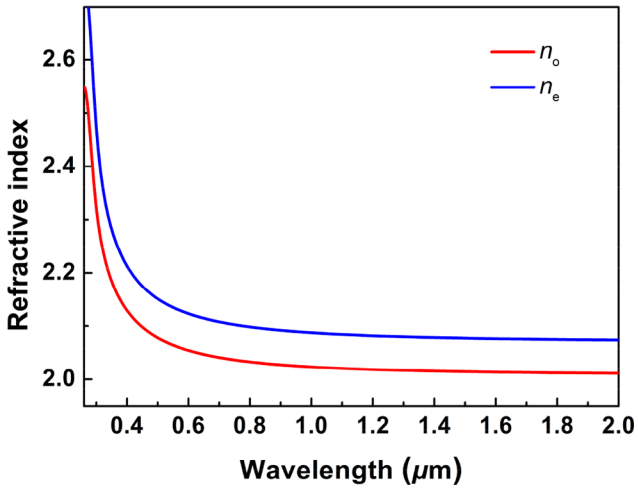

Figure 8. Calculated refractive index dispersion curves for the Li_2ZrTeO_6 crystal.

The curves exhibit anisotropy and comply with the order of $n_0 < n_e$, indicating that Li_2ZrTeO_6 is a positive uniaxial crystal with a moderate birefringence of $\Delta n = 0.0638$ at 1064 nm. The relatively large birefringence of Li_2ZrTeO_6 at 1064 nm is fairly close to the value of $LiNbO_3$ ($\Delta n = 0.0786$)[72] and is consistent with the experimentally phase-matchable behavior.

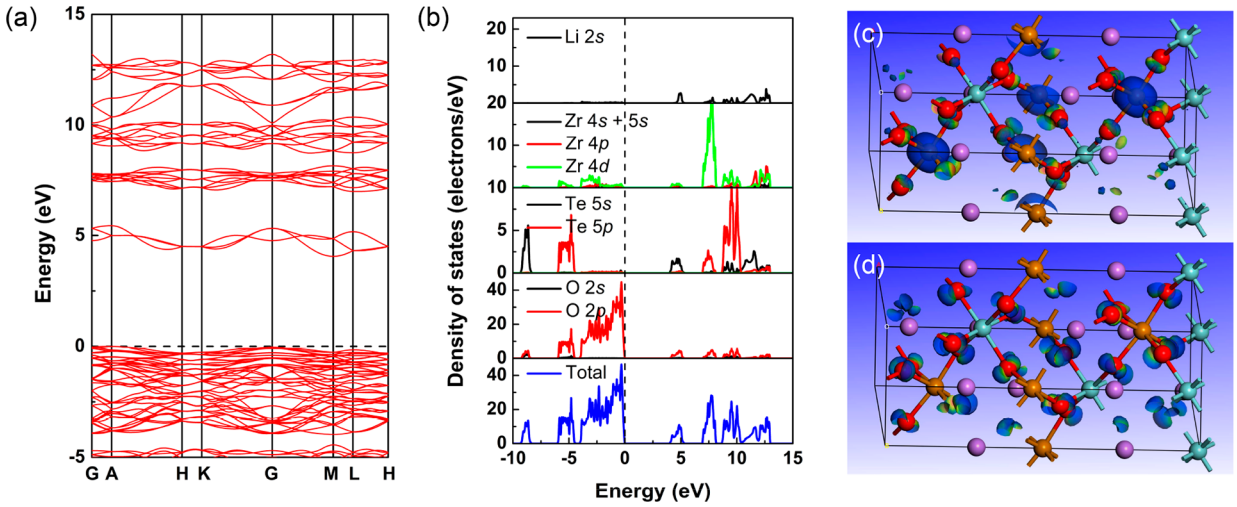

Figure 7. Electronic structure of Li_2ZrTeO_6. (a) Energy band structure. (b) TDOS and PDOS curves. The Fermi level is set at 0.0 eV (dashed vertical line). (c and d) Calculated LUMO and HOMO in the CB and VB, respectively.

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CONCLUSION

In summary, we have rationally designed a new tellurate crystal, Li_2ZrTeO_6, by the element substitution of Nb for Zr and Te from $LiNbO_3$. This tellurate features an octahedral framework geometry that inherits the structural merits of $LiNbO_3$. As a result, Li_2ZrTeO_6 displays the optical properties required for NLO applications, involving noncentrosymmetric crystal structure, moderate birefringence, and phase-matchability. Meanwhile, Li_2ZrTeO_6 achieves more outstanding optical damage resistant behavior exceeding 22 times that of $LiNbO_3$, which is more suitable for high-energy NLO applications. Furthermore, Li_2ZrTeO_6 exhibits the widest IR absorption edge (7.4 μm) among all of the noncentrosymmetric tellurates reported so far. These excellent properties make Li_2ZrTeO_6 an attractive candidate for favorable NLO applications. The substitution of Nb for Zr and Te from $LiNbO_3$ sheds light on a versatile opportunity to design new NLO crystals with high performance.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/jacs.8b08803.

Additional figures and tables (PDF)

X-ray crystallographic data (CIF)

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Notes

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